

Bell-Scale Growth of OEIS A088714

A rigorous n th-root asymptotic, quantitative bounds,
and a separately identified finer conjecture

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Abstract

Let $A(x) = \sum_{n \geq 0} a_n x^n$ be the unique formal power series satisfying $A(x) = 1 + xA(x)^2A(xA(x))$. Its coefficients form OEIS A088714. We prove

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right), \quad a_n^{1/n} \sim \frac{n}{e \log n}.$$

The proof is based on an exact triangular recurrence obtained from composition iterates. A simple path through that recurrence supplies the lower bound. The upper bound comes from a coefficientwise inequality for shifted Poisson moments, proved here by a positivity argument involving Stirling numbers. More explicitly, for every $n \geq 1$,

$$\max_{1 \leq h \leq n+1} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda),$$

where T_n is the n th Touchard polynomial. Consequently the ordinary generating function has radius zero, whereas the exponential generating function is entire. Exact coefficient computations accompany the proof. They suggest a considerably sharper Bell-number comparison, but that comparison and its multiplicative constant are not claimed as established results.

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1 The problem and the precise result

The OEIS entry A088714 defines the sequence through the formal functional equation [1]

$$A(x) = 1 + xA(x)^2A(xA(x)), \quad A(0) = 1. \quad (1.1)$$

Its beginning is

$$1, 1, 3, 13, 69, 419, 2809, 20353, 157199, 1281993, 10963825, \dots$$

The index convention throughout is $a_0 = 1$, $a_1 = 1$, $a_2 = 3$. All logarithms are natural. Formal composition is legitimate because the series substituted into A has zero constant term.

The composition in (1.1) makes this a different problem from ordinary algebraic generating-function asymptotics. In fact, the answer rules out a positive radius of convergence. One must not assume that the formal identity initially defines a holomorphic function near the origin.

Theorem 1.1 (Main asymptotic). *For the coefficients of (1.1),*

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (1.2)$$

Equivalently, at the level of n th-root asymptotics,

$$a_n^{1/n} = \frac{n}{e \log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right), \quad \lim_{n \rightarrow \infty} \frac{e \log n}{n} a_n^{1/n} = 1. \quad (1.3)$$

In particular,

$$a_n = \left(\frac{n}{e \log n}\right)^n \exp\left(O\left(\frac{n \log \log n}{\log n}\right)\right). \quad (1.4)$$

The $-n$ in (1.2) is important: it identifies the constant $1/e$ in (1.3). The weaker assertion $\log a_n \sim n \log n$ would miss both the logarithmic denominator and this constant.

What the theorem does not say. Equation (1.4) is not the multiplicative equivalent $a_n \sim (n/(e \log n))^n$. The error in its logarithm is $o(n)$, but it need not tend to zero. Likewise, an n th-root equivalent does not by itself imply an equivalent for a_n/a_{n-1} . Section 8 discusses a finer numerical pattern without confusing it with the theorem.

A useful feature of the argument is that it gives bounds for every index, not merely an eventual limiting statement.

Theorem 1.2 (Finite-index comparison). *Define the Touchard polynomials by*

$$T_n(\lambda) = \sum_{j=0}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \lambda^j, \quad (1.5)$$

where $\left\{ \begin{matrix} n \\ j \end{matrix} \right\}$ denotes a Stirling number of the second kind, and let $T'_n(\lambda)$ mean differentiation with respect to λ . For every integer $n \geq 1$,

$$\max_{h \in \{1, \dots, n+1\}} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (1.6)$$

For every $r > 0$ and $\lambda > 0$ this implies the entirely explicit bound

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n \frac{n!}{r^n} (e^r - 1) \exp(\lambda(e^r - 1)). \quad (1.7)$$

The remainder of the proof is self-contained. The OEIS supplies the sequence identification and related formal identities. Standard Bell number and Lambert W information is used for interpretation and numerical comparison, not to assume the answer for A088714.

2 Formal identities and the exact coefficient recurrence

2.1 Existence, uniqueness, and direct extraction

In (1.1), the coefficient of x^n on the right depends only on $a_0, \dots, a_{n-1}$. Starting with $a_0 = 1$ therefore determines the coefficients successively and proves existence and uniqueness in $\mathbb{Z}[[x]]$. The same observation proves their nonnegativity. Expanding the outer occurrence of A gives the exact identity

$$a_n = \sum_{k=0}^{n-1} a_k [x^{n-1-k}] A(x)^{k+2}, \quad n \geq 1. \quad (2.1)$$

This recurrence is useful for independently checking computed coefficients. It also explains the finer heuristic in Section 8, but it is not the most convenient recurrence for our rigorous upper bound.

2.2 Passage to composition iterates

Put $B(x) = xA(x)$. Equation (1.1) becomes

$$B(x) = x + B(x)B(B(x)). \quad (2.2)$$

Since $B(x) = x + O(x^2)$, it has a formal compositional inverse. Substitution of that inverse into (2.2) yields

$$B^{(-1)}(x) = x - xB(x) = x - x^2A(x). \quad (2.3)$$

This is the reversion identity also recorded in the OEIS entry [1].

For each integer $k \geq 0$, define

$$F_k(x) = \frac{B^{\circ k}(x)}{x}, \quad F_0(x) = 1, \quad F_1(x) = A(x). \quad (2.4)$$

Here $B^{\circ 0}(x) = x$ and $B^{\circ k}$ denotes k -fold composition, not an ordinary power. Composing (2.2) with $B^{\circ(k-1)}$ and dividing by x gives

$$F_k(x) = F_{k-1}(x) + xF_k(x)F_{k+1}(x), \quad k \geq 1. \quad (2.5)$$

The first few equations are the infinite system displayed in the OEIS entry; the derivation here fixes the indexing and proves the general identity [1].

Write

$$F_k(x) = \sum_{n \geq 0} c_{n,k} x^n.$$

Then

$$c_{0,k} = 1 \quad (k \geq 0), \quad c_{n,0} = 0 \quad (n \geq 1), \quad a_n = c_{n,1}. \quad (2.6)$$

Coefficient extraction in (2.5) gives

$$c_{n,k} - c_{n,k-1} = \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1}. \quad (2.7)$$

Equivalently,

$$c_{n,k} = \sum_{j=1}^k \sum_{i=0}^{n-1} c_{i,j} c_{n-1-i,j+1}. \quad (2.8)$$

Every term on the right has first index less than n . Consequently this is a genuine triangular recurrence despite the occurrence of $k + 1$. It proves positivity and monotonicity in k , and is the main structural tool of the article.

For illustration, the first three coefficient polynomials are

$$\begin{aligned} c_{1,k} &= k, \\ c_{2,k} &= k^2 + 2k, \\ c_{3,k} &= k^3 + 5k^2 + 7k. \end{aligned}$$

Evaluating at $k = 1$ gives 1, 3, 13 as required. These are not ordinary Touchard polynomials; the Poisson polynomials introduced later are comparison objects, not an exact formula for the sequence.

2.3 Relation to the signed reversion sequence

For completeness, set

$$g(x) = -B^{(-1)}(-x) = x + x^2 A(-x).$$

The inverse identity implies

$$g(g(x)) = (1 + x)g(x).$$

For $n \geq 2$ its coefficient of x^n is $(-1)^{n-2} a_{n-2}$. Thus the signed series in A067145 has the same coefficient magnitudes, with this explicit index shift [2]. This identity is an additional verification route, not an analytic-convergence argument.

3 A lower bound from an increasing-index path

Lemma 3.1. *For every $n \geq 0$ and every integer $k \geq 1$,*

$$c_{n,k} \geq k^n. \quad (3.1)$$

Proof. The case $n = 0$ is immediate. Suppose the statement holds for smaller first indices. By (2.8),

$$\begin{aligned} c_{n,k} &\geq \sum_{j=1}^k \sum_{i=0}^{n-1} j^i (j+1)^{n-1-i} \\ &= \sum_{j=1}^k ((j+1)^n - j^n) = (k+1)^n - 1 \geq k^n. \end{aligned}$$

The inner sum is a difference of two n th powers, and the outer sum then telescopes. This completes the induction. $\square$

The $i = 0$ summand of (2.7) and positivity imply

$$c_{n,k} \geq c_{n-1,k+1} \quad (n \geq 1).$$

Iterating this inequality $h - 1$ times, for $1 \leq h \leq n + 1$, gives

$$a_n = c_{n,1} \geq c_{n-h+1,h} \geq h^{n-h+1}. \quad (3.2)$$

This proves the lower half of Theorem 1.2.

3.1 Optimizing the lower bound

Let $N = n + 1$ and temporarily allow h to be real. The function

$$\phi_N(h) = (N - h) \log h$$

has derivatives

$$\phi'_N(h) = \frac{N}{h} - 1 - \log h, \quad \phi''_N(h) = -\frac{N}{h^2} - \frac{1}{h} < 0.$$

Its unique maximum is attained at

$$h_*(1 + \log h_*) = N, \quad h_* = \frac{N}{W(eN)}. \quad (3.3)$$

The positive real branch of Lambert W , characterized by $W(z)e^{W(z)} = z$, is intended [4]. The exact maximizing integer is among $\lfloor h_* \rfloor$ and $\lceil h_* \rceil$. If $w_* = W(eN)$, the continuous maximum is

$$\phi_N(h_*) = N \left(w_* - 2 + \frac{1}{w_*} \right). \quad (3.4)$$

Rounding to a nearest integer changes this value by $O((\log N)^2/N)$: on an interval of fixed length around h_* , $|\phi''_N(h)| = O((\log N)^2/N)$, and $\phi'_N(h_*) = 0$. This provides an accurate, explicit lower bound even when the leading logarithmic expression is numerically crude.

For the main theorem, a simpler choice suffices. Put

$$L = \log n, \quad \ell = \log \log n, \quad h = \left\lfloor \frac{n}{L} \right\rfloor.$$

As $n \rightarrow \infty$,

$$\log h = L - \ell + O(L/n), \quad h = n/L + O(1).$$

Hence (3.2) yields

$$\log a_n \geq n(L - \ell - 1) + \frac{n\ell}{L} - O(L). \quad (3.5)$$

In particular the lower bound has the same $n \log n$, $-n \log \log n$, and $-n$ terms as the claimed answer.

4 The Poisson-moment comparison inequality

The upper bound requires a comparison family that respects the increase from k to $k + 1$ in (2.7). Ordinary exponential bounds are too small, while a rough factorial bound loses the logarithmic factor. Shifted Poisson moments provide the needed intermediate scale.

Let X have the Poisson distribution with mean $\lambda > 0$, and define

$$M_n(k) = \mathbb{E}(k + X)^n, \quad T_n(\lambda) = M_n(0). \quad (4.1)$$

Their exponential generating functions are

$$\sum_{n \geq 0} M_n(k) \frac{z^n}{n!} = \exp(kz + \lambda(e^z - 1)), \quad \sum_{n \geq 0} T_n(\lambda) \frac{z^n}{n!} = \exp(\lambda(e^z - 1)). \quad (4.2)$$

Expanding the last exponential proves (1.5); at $\lambda = 1$ these polynomials give the Bell numbers. Standard Bell identities are recorded in [3]; the identities needed below also follow directly from (4.2).

For $n \geq 1$ set

$$D_n(k) = M_n(k) - M_n(k-1), \quad S_n(k) = \sum_{i=0}^{n-1} M_i(k) M_{n-1-i}(k+1). \quad (4.3)$$

The central estimate is the following.

Lemma 4.1 (Shifted Poisson convolution inequality). *For every $n \geq 1$, $k \geq 1$, and $\lambda > 0$,*

$$S_n(k) \leq \left(1 + \frac{2}{\lambda}\right) D_n(k). \quad (4.4)$$

In fact, after multiplication by λ , the difference between the right and left sides is a polynomial with nonnegative coefficients in λ and $k-1$.

4.1 Reduction to the shift $k = 1$

Write $u = k - 1$. The binomial theorem gives

$$D_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} D_r(1). \quad (4.5)$$

There is a matching identity for the ordinary convolution:

$$S_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} S_r(1). \quad (4.6)$$

To verify it, expand both shifted moments in the definition of $S_n(k)$. For fixed nonnegative integers p, q with $p + q = r - 1$, the coefficient sum over the convolution index is

$$\sum_i \binom{i}{p} \binom{n-1-i}{q} = \binom{n}{p+q+1} = \binom{n}{r}.$$

This identity counts a subset of r elements of an n -element ordered set by the location of its $(p+1)$ st element. Thus it holds including all boundary cases. Equations (4.5) and (4.6) reduce the lemma to $k = 1$.

Differentiating (4.2) with respect to λ shows

$$D_n(1) = T'_n(\lambda). \quad (4.7)$$

Suppress the argument λ temporarily, write $S_n = S_n(1)$, and set

$$H_n = (\lambda + 2)T'_n - \lambda S_n. \quad (4.8)$$

We shall prove that H_n has nonnegative coefficients for every $n \geq 1$.

4.2 A recurrence for the difference polynomial

The moment identities imply

$$T_{n+1} = \lambda(T_n + T'_n), \quad (4.9)$$

$$M_{n+1}(k) = (k + \lambda)M_n(k) + \lambda M'_n(k). \quad (4.10)$$

Here and below a prime means differentiation in λ , not in k . Applying (4.10) to both factors in S_n gives

$$S_{n+1} = \frac{\lambda}{2}S'_n + \left(\lambda + \frac{3}{2}\right)S_n + \frac{1}{2}(M_n(1) + M_n(2)). \quad (4.11)$$

Indeed, the two shifted convolution sums add up to $2S_{n+1} - M_n(1) - M_n(2)$. Furthermore,

$$M_n(1) = T_n + T'_n, \quad M_n(2) = T_n + 2T'_n + T''_n.$$

Substitute these formulas and (4.9) into (4.8). A direct simplification gives the exact recurrence

$$H_{n+1} = \frac{\lambda}{2}H'_n + (\lambda + 1)H_n + K_n, \quad (4.12)$$

where

$$K_n = \frac{\lambda(\lambda + 1)}{2}T''_n - 2\lambda T'_n + 2T_n. \quad (4.13)$$

The initial difference polynomials are

$$\begin{aligned} H_1 &= 2, \\ H_2 &= 2 + 2\lambda, \\ H_3 &= 2 + 6\lambda + \lambda^2, \\ H_4 &= 2 + 14\lambda + 8\lambda^2. \end{aligned}$$

They all have nonnegative coefficients.

4.3 Why the forcing polynomial is nonnegative

From (1.5), the coefficient of λ^j in K_n is

$$[\lambda^j]K_n = \frac{(j-1)(j-4)}{2} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} + \frac{j(j+1)}{2} \left\{ \begin{matrix} n \\ j+1 \end{matrix} \right\}. \quad (4.14)$$

For $n \geq 4$, this is visibly nonnegative except possibly when $j = 2$ or $j = 3$. Those cases require only

$$\left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} \leq 3 \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \quad (n \geq 3), \quad \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \leq 6 \left\{ \begin{matrix} n \\ 4 \end{matrix} \right\} \quad (n \geq 4). \quad (4.15)$$

Here is a common elementary proof. If a partition of n elements into r blocks has $n > r$, it has a nonsingleton block. Choose such a block by any fixed deterministic rule and split off a specified element of it. Remember the unordered pair of blocks created. Merging that pair recovers the original partition. This is an injection into partitions with $r + 1$ blocks together with a choice of two of those blocks. Therefore

$$\left\{ \begin{matrix} n \\ r \end{matrix} \right\} \leq \binom{r+1}{2} \left\{ \begin{matrix} n \\ r+1 \end{matrix} \right\} \quad (n > r). \quad (4.16)$$

Taking $r = 2, 3$ proves (4.15). Thus K_n has nonnegative coefficients for all $n \geq 4$.

Starting from H_4 , recurrence (4.12) preserves coefficientwise nonnegativity: differentiation, multiplication by $\lambda/2$ or by $\lambda + 1$, and addition of K_n all preserve it. Together with the three earlier base cases, this proves $H_n \geq 0$ coefficientwise for all $n \geq 1$. By (4.8) and (4.7),

$$\lambda S_n(1) \leq (\lambda + 2)D_n(1).$$

Finally, (4.5)–(4.6) extend the result to all $k \geq 1$, proving Lemma 4.1.

5 The global upper bound and completion of the proof

5.1 Comparison with the triangular recurrence

Fix any $\lambda > 0$ and write

$$\gamma = 1 + \frac{2}{\lambda}.$$

We claim that, simultaneously for all $n \geq 0$ and $k \geq 0$,

$$c_{n,k} \leq \gamma^n M_n(k). \quad (5.1)$$

For $n = 0$, both sides equal 1. Assume the bound for smaller first indices. Equations (2.8) and (4.4) imply, for $n \geq 1$,

$$\begin{aligned} c_{n,k} &\leq \gamma^{n-1} \sum_{j=1}^k S_n(j) \\ &\leq \gamma^n \sum_{j=1}^k (M_n(j) - M_n(j-1)) \\ &= \gamma^n (M_n(k) - M_n(0)) \leq \gamma^n M_n(k). \end{aligned} \quad (5.2)$$

The boundary $k = 0$ is immediate. This proves the induction.

Importantly, the penultimate expression is sharper than the induction claim. At $k = 1$, (4.7) yields

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (5.3)$$

Because λ was arbitrary, this proves the upper half of Theorem 1.2.

Differentiation of (4.2) gives

$$\sum_{n \geq 0} T'_n(\lambda) \frac{r^n}{n!} = (e^r - 1) \exp(\lambda(e^r - 1)).$$

All coefficients are nonnegative. Any one term is bounded by the whole sum when $r > 0$, and (1.7) follows. This argument is just a positive-coefficient bound; it does not require any analyticity of $A(x)$.

5.2 An explicit parameter choice

Let $L = \log n$ and $\ell = \log L$ as before. For all sufficiently large n , take

$$\lambda = L, \quad r = L - 2\ell > 0. \quad (5.4)$$

Then $e^r = n/L^2$. Taking logarithms in (1.7), and using $\log(e^r - 1) \leq r$, gives

$$\begin{aligned} \log a_n &\leq \log n! - n \log r + n \log(1 + 2/L) + L(e^r - 1) + r \\ &= \log n! - n \log(L - 2\ell) + n \log(1 + 2/L) + \frac{n}{L} - 2\ell. \end{aligned} \quad (5.5)$$

The elementary estimates

$$\begin{aligned} \log n! &= nL - n + O(L), \\ \log(L - 2\ell) &= \ell - \frac{2\ell}{L} + O\left(\frac{\ell^2}{L^2}\right), \\ \log(1 + 2/L) &= \frac{2}{L} + O(L^{-2}) \end{aligned}$$

now show

$$\log a_n \leq n(L - \ell - 1) + \frac{2n\ell + 3n}{L} + O\left(\frac{n\ell^2}{L^2} + L\right). \quad (5.6)$$

For the first estimate, even Stirling's full formula is unnecessary: bounding the sum $\sum_{j=1}^n \log j$ between its neighboring integrals already gives $\log n! = n \log n - n + O(\log n)$.

The lower estimate (3.5) and upper estimate (5.6) together give (1.2). Dividing by n and exponentiating proves (1.3) and (1.4). This completes the proof of Theorem 1.1.

Remark 5.1 (Why a tunable comparison is necessary). A bound of the form $a_n \leq C^n n!$ cannot identify the factor $(\log n)^{-n}$. Even a comparison with C^n times a fixed Bell-type sequence leaves an unwanted constant C in the n th root. Here the factor $1 + 2/\lambda$ can approach 1, and the valid choice $\lambda = \log n$ makes its total logarithmic cost only $O(n/\log n)$. The dependence of the Poisson mean on n is fully allowed because (5.3) holds for every positive λ at every index.

6 Consequences and comparison with Bell numbers

6.1 Superexponential but subfactorial growth

The main theorem immediately implies

$$\frac{a_n}{C^n} \longrightarrow \infty \quad \text{for every fixed } C > 0. \quad (6.1)$$

On the other hand,

$$\left(\frac{a_n}{n!}\right)^{1/n} = \frac{1}{\log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right) \longrightarrow 0. \quad (6.2)$$

Thus $a_n/(C^n n!) \rightarrow 0$ for every fixed $C > 0$. These two assertions locate the growth more precisely than the phrase “almost factorial”: there is a factor of approximately $(\log n)^n$ between $n!$ and a_n on the logarithmic scale.

For every fixed $s < 1$ and $C > 0$,

$$\frac{a_n}{C^n (n!)^s} \longrightarrow \infty,$$

because the logarithm of this quotient has dominant term $(1-s)n \log n$. In the convention that Gevrey order s means a bound by $CK^n(n!)^s$, the series has exact Gevrey order 1, with zero factorial exponential type. This statement specifies the convention, as Gevrey indexing is not uniform across the literature.

6.2 The two generating functions behave differently

By the Cauchy–Hadamard formula,

$$\text{rad } A = \frac{1}{\limsup_{n \geq 0} a_n^{1/n}} = 0.$$

In particular, there is no holomorphic germ at zero whose Taylor series has these coefficients and satisfies (1.1). The formal series itself remains perfectly well-defined and unique.

In contrast, the exponential generating function

$$\mathcal{A}(t) = \sum_{n \geq 0} a_n \frac{t^n}{n!} \quad (6.3)$$

is entire by (6.2). Summing the finite-index upper bound also gives, for every $t \geq 0$ and $\lambda > 0$,

$$\mathcal{A}(t) \leq 1 + (e^{\gamma t} - 1) \exp(\lambda(e^{\gamma t} - 1)), \quad \gamma = 1 + 2/\lambda. \quad (6.4)$$

Corollary 6.1 (Double-exponential scale). *Along the positive real axis,*

$$\lim_{t \rightarrow \infty} \frac{\log \log \mathcal{A}(t)}{t} = 1. \quad (6.5)$$

Proof. For a fixed λ , (6.4) bounds the limit superior by $1 + 2/\lambda$. Letting $\lambda \rightarrow \infty$ gives the upper bound 1. For the lower bound fix $0 < \varepsilon < 1$ and take $n = \lfloor e^{(1-\varepsilon)t} \rfloor$. The single term $a_n t^n / n!$ and Theorem 1.1 give

$$\log \left(\frac{a_n t^n}{n!} \right) = n(\log t - \log \log n + o(1)) = n(-\log(1 - \varepsilon) + o(1)).$$

The constant inside the last parentheses is positive. Taking a further logarithm shows that the limit inferior in (6.5) is at least $1 - \varepsilon$. Let $\varepsilon \downarrow 0$. $\square$

Since the coefficients of $\mathcal{A}$ are nonnegative, its maximum modulus on $|z| = t$ equals $\mathcal{A}(t)$. Consequently its usual entire function order is infinite. Equation (6.5), rather than just “infinite order,” states the useful growth scale.

6.3 Bell-number scale is not Bell-number equivalence

Let $B_n = T_n(1)$ denote the Bell numbers. A standard asymptotic, with $w = W(n)$, is [3]

$$B_n \sim \frac{\exp(n(w - 1 + 1/w) - 1)}{\sqrt{1 + w}}. \quad (6.6)$$

In particular $B_n^{1/n} \sim n/(e \log n)$. Therefore our theorem implies

$$\left(\frac{a_n}{B_n} \right)^{1/n} \longrightarrow 1, \quad a_n = B_n \exp(o(n)). \quad (6.7)$$

These are the senses in which the sequence has Bell-number-scale growth. They do not assert that a_n/B_n is bounded or converges. Indeed, the finite computations in the next section show a substantial and growing discrepancy. The formula (6.6) is a comparison formula for Bell numbers, not a formula silently transferred to A088714.

7 Exact computation and numerical diagnostics

7.1 A triangular dynamic program

To obtain $a_0, \dots, a_N$, store $c_{0,k} = 1$ for $0 \leq k \leq N + 1$. For each $n = 1, \dots, N$, compute

$$c_{n,0} = 0, \quad c_{n,k} = c_{n,k-1} + \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1} \quad (1 \leq k \leq N - n + 1).$$

Then output $a_n = c_{n,1}$. The triangular cutoff is sufficient: every requested entry on the right was computed at an earlier first index. The number of products is exactly

$$\sum_{n=1}^N n(N - n + 1) = \frac{N(N + 1)(N + 2)}{6}.$$

Thus the algorithm uses $O(N^3)$ arbitrary-precision arithmetic operations and $O(N^2)$ integer cells. These are arithmetic counts, not unit-cost bit-complexity claims; the integers grow with N .

The archive contains both a dependency-free Python implementation and a faster C++ implementation using GMP. The supplied data were computed with exact integer arithmetic through $n = 1200$. The Python and C++ outputs agree through $n = 400$. Independent truncated-polynomial checks verify (1.1) and (2.3); the supplied verification script also checks the auxiliary polynomial recurrences and finite bounds on explicitly reported finite ranges.

7.2 What the diagnostics measure

Define

$$Q_n = \frac{e \log n}{n} a_n^{1/n}, \quad (7.1)$$

$$R_n = \frac{a_n}{a_{n-1}}, \quad (7.2)$$

$$D_n^{\text{num}} = \log \left(\frac{a_n}{B_n} \right) - W(n)^2 - 3W(n). \quad (7.3)$$

The notation D_n^{num} is distinct from the Poisson moment difference $D_n(k)$ in Section 4. The main theorem proves $Q_n \rightarrow 1$. The quantities R_n and D_n^{num} examine finer behavior not established by that theorem.

n	$\log_{10} a_n$	Q_n	$R_n - n/W(n)$	D_n^{num}
20	17.180354	2.942966	2.650843	-3.494768
50	53.134060	2.457011	2.312570	-3.262262
100	123.688846	2.159875	2.205968	-3.172516
200	286.269488	1.944240	2.146342	-3.115335
400	657.374892	1.791450	2.110948	-3.078375
800	1495.731414	1.682461	2.089104	-3.054662
1200	2408.704723	1.633120	2.080197	-3.045054

Table 1: Diagnostics from independently generated exact coefficients. Only the limit $Q_n \rightarrow 1$ is proved here. The final column tests the conjecture of Section 8.

The relatively slow approach of Q_n to 1 is not surprising: $\log \log n / \log n$ is still appreciable at these indices. Substituting only the first term $(n/(e \log n))^n$ is therefore not a numerically accurate

approximation to a_n , despite its correct n th-root scale. The optimized finite lower and upper bounds offer certified alternatives when actual error control is required.

The normalization in (7.3) is motivated next. Its numerical behavior is evidence for a conjecture, not evidence that upgrades Theorem 1.1 to a multiplicative equivalent. In particular no rigorous enclosure for a limiting constant is extracted from the table.

8 A finer asymptotic: formal motivation and its missing step

8.1 The leading endpoint balance

Write $R_n = a_n/a_{n-1}$ and consider (2.1). Near its outer endpoint put $k = n - 1 - m$. Then the corresponding summand is

$$a_{n-1-m}[x^m]A(x)^{n+1-m}.$$

Since $a_1 = 1$, the first approximation to the coefficient is $n^m/m!$. If neighboring ratios can be treated as locally constant, the endpoint sum divided by a_n is approximately

$$\frac{1}{R_n} \sum_{m \geq 0} \frac{(n/R_n)^m}{m!} = \frac{e^{n/R_n}}{R_n}.$$

Setting this expression equal to 1 leads to

$$R_n \log R_n \approx n, \quad R_n \approx \frac{n}{W(n)}.$$

This explains the natural Lambert W scale of the ratios. It is not a proof: local ratio regularity and uniform control of the other summands have been assumed in deriving the approximation.

8.2 The first correction and a candidate normalization

There is more information in the same calculation. From the first few coefficients,

$$\log A(x) = x + \frac{5}{2}x^2 + O(x^3).$$

Expanding the near-endpoint coefficient for m of logarithmic size gives the formal local approximation

$$[x^m]A(x)^{n+1-m} = \frac{n^m}{m!} \left(1 + \frac{3m(m-1)}{2n} + \dots \right). \quad (8.1)$$

The factor $3/2$ combines the quadratic term of $\log A$ with the replacement of $n+1-m$ by n .

Suppose, for this calculation, that the ratio has a smooth expansion near $R_n = n/w$, where $w = W(n)$. Put

$$t = n/R_n, \quad d = \frac{w}{1+w}.$$

The local variation of the ratios contributes $m(m+1)d/(2n)$ to the bracket multiplying the Poisson kernel. Averaging $m(m-1)$ and $m(m+1)$ against that kernel replaces them by t^2 and $t^2 + 2t$, respectively. The opposite endpoint, where one inner factor has almost all the degree, begins with $2a_{n-1}$ from $A(x)^2$. At this formal order the equation becomes

$$1 \approx \frac{e^t}{R_n} \left(1 + \frac{(3+d)t^2 + 2dt}{2n} \right) + \frac{2}{R_n}. \quad (8.2)$$

Writing $R_n = n/w + \delta$ and expanding gives

$$\delta(1+w) \approx 2 + \frac{(3+d)w + 2d}{2},$$

and hence the candidate ratio approximation

$$R_n \approx \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2}. \quad (8.3)$$

This is consistent with the exact data, but the approximation signs in this subsection are intentional.

A normalization whose successive quotient has these same first terms is

$$\mathcal{N}_n = \frac{\exp(n(w-1+1/w) - 1 + w^2 + 3w)}{\sqrt{1+w}}, \quad w = W(n). \quad (8.4)$$

Indeed, using $w' = w/(n(1+w))$ and Taylor expansion in the real variable n shows

$$\frac{\mathcal{N}_n}{\mathcal{N}_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^3}{n}\right). \quad (8.5)$$

One way to check the differentiation is to note that

$$\frac{d}{dn} [n(w-1+1/w)] = w.$$

The additional derivative from $w^2 + 3w - \frac{1}{2} \log(1+w)$ supplies the constant and rational correction in (8.5).

Conjecture 8.1 (Finer Bell comparison). *There exists a constant $C > 0$ such that, with $w = W(n)$,*

$$a_n \sim C B_n \exp(w^2 + 3w) \sim C \mathcal{N}_n. \quad (8.6)$$

The existence or numerical value of C is not established in this article. The data and the endpoint calculation motivate the displayed form, rather than merely an arbitrary fit to an exponential correction.

8.3 A precise sufficient condition for the conjecture

The unresolved analytic step can be stated cleanly.

Proposition 8.2 (Conditional implication). *Suppose that for some fixed $p \geq 0$ the actual ratios satisfy*

$$\frac{a_n}{a_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^p}{n}\right). \quad (8.7)$$

Then Conjecture 8.1 follows with some finite, strictly positive constant C .

Proof. Compare (8.7) with (8.5). Since both ratios are asymptotic to $n/W(n)$, their quotient is

$$1 + O\left(\frac{W(n)^{\max(p,3)+1}}{n^2}\right).$$

The error is summable because $W(n) = O(\log n)$. Therefore the successive differences of $\log(a_n/\mathcal{N}_n)$ are absolutely summable. This logarithm converges to a finite real number, so $a_n/\mathcal{N}_n$ tends to a finite positive constant. Formula (6.6) then gives the Bell-number version. $\square$

The main theorem does not supply (8.7). In particular, it does not justify replacing all nearby coefficient ratios by a smooth function, nor does it control every non-endpoint contribution at a summable error level. A proof of such uniform endpoint estimates would be a genuine strengthening of the present result. This identifies the remaining task without presenting a formal asymptotic calculation as a completed proof.

9 Reproducibility and verification scope

The archive is organized so that the mathematical theorem does not depend on a numerical black box. The principal files are `article.tex`, `article.pdf`, the exact data file `data/coefficients.txt`, and the scripts under `code/`.

The dependency-free command

```
python3 code/verify.py
```

checks the initial OEIS terms; verifies the defining composition and inverse identities by independent truncated-polynomial arithmetic; checks the Poisson difference-polynomial identity and coefficient signs on a finite range; and tests both sides of the finite-index bound using exact integer arithmetic. Its actual output is supplied in `data/verification.txt`. These are useful implementation audits; the all-index positivity proof is the argument in Section 4, not an extrapolation from those finite checks.

The command

```
python3 code/coefficients.py -n 400 -output coefficients.txt
```

recomputes the first 401 terms without third-party Python packages. For a larger table, `code/compute_gmp.cpp` provides the same recurrence in C++ with GMP. Instructions in `README.md` describe compilation and analysis. The diagnostic calculations use exact integers as input and arbitrary-precision floating-point logarithms only at the presentation stage; no computed coefficient is rounded.

No computer-assisted proof of the finer conjecture is claimed. The article's rigorous conclusion is Theorem 1.1, supported by the explicit all-index inequalities of Theorem 1.2.

10 Conclusion

OEIS A088714 has the precise n th-root growth

$$a_n^{1/n} \sim \frac{n}{e \log n}.$$

The accompanying logarithmic error estimate is

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right).$$

The mechanism is visible in the proof. Composition iterates turn the functional equation into a positive triangular recurrence. Moving along its increasing-index direction gives roughly h^{n-h} , whose optimum occurs at h of order $n/\log n$. Shifted Poisson moments furnish a compatible upper comparison and recover the same leading scale, including the factor $1/e$.

The result determines the growth at logarithmic precision $o(n)$ and settles the convergence classification of both generating functions. The finer proposed factor $\exp(W(n)^2 + 3W(n))$ remains explicitly separate from the proved theorem.

References

- [1] The OEIS Foundation Inc., *A088714: G.f. satisfies $A(x) = 1 + xA(x)^2A(xA(x))$* , entry attributed to Paul D. Hanna, with subsequent contributions. <https://oeis.org/A088714>. Text form consulted: <https://oeis.org/search?q=id:A088714&fmt=text>. Accessed September 20, 2026.
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